Supplementary File
Post-quantum Secure Spatial Dataset Joinable Search Processing

November 2026

1 Overview

This supplementary file provides additional experimental results and analyses that could not be fully included in the main manuscript due to space limitations. The supplementary materials are intended to provide further details and empirical evidence for the proposed PQJS scheme, including comprehensive dataset statistics, Bloom filter false-positive analysis, detailed time breakdowns, scalability and updateability analysis, and visualization of representative spatial search results. These additional results complement the experimental evaluation presented in the main manuscript and provide further insights into the effectiveness, efficiency, scalability, and practical applicability of PQJS. Due to the large size of the vector graphics rendered for Fig. 1 and Fig. 3 and compatibility issues across different document rendering environments, non-vector versions are provided for these figures in the supplementary file. The original vector versions of Fig. 1 and Fig. 3 are available in the accompanying GitHub repository.

2 Baseline Comparison Details

Comparison with PSDJS/PSDJS+ [1]. PSDJS relies on EASPE-based matrix encryption, which introduces limitations in key management and key update. Moreover, PSDJS vectorizes each spatial grid for joinable search, so its approximation error in joinability evaluation primarily stems from the vectorization process itself, which also incurs additional computational overhead. To address this efficiency concern, the enhanced variant PSDJS+ introduces a two-layer index combining spatial grids and HNSW for candidate pruning. However, since HNSW retrieval is approximate by design, PSDJS+ improves efficiency at the cost of search accuracy. More fundamentally, both schemes formalize the joinable search problem solely in terms of spatial coverage, neglecting the contribution of individual data points within each dataset. In contrast, PQJS explicitly incorporates spatial data-point density into the problem formulation, enabling a more comprehensive evaluation of joinability. The search results of PQJS under this formulation are reported and made publicly available in the accompanying GitHub repository. To achieve acceptable search efficiency under CKKS-based homomorphic encryption, PQJS introduces a GMBR-based pre-filtering stage that prunes over 99% of candidate datasets, making the scheme practically deployable. Nevertheless, since PSDJS and PSDJS+ operate under a different joinability formulation, a direct comparison beyond search time and storage overhead is not applicable.

Comparison with OJSP/CJSP [2]. Yang et al. decompose the spatial dataset joinable search problem into two sub-problems: Overlap Joinable Search Processing (OJSP) and Coverage Joinable Search Processing (CJSP). Both sub-problems share a common first stage that identifies datasets joinable with the query by directly checking MBR intersection, which may introduce false positives. OJSP then employs an R-tree for

[1] Zhang et al. A Privacy-preserving Spatial Dataset Joinable Search in Cloud. Proceedings of the 34th ACM International Conference on Information and Knowledge Management. url: https://doi.org/10.1145/3746252.3761218

[2] Yang et al. Joinable Search over Multi-Source Spatial Datasets: Overlap, Coverage, and Efficiency. Proceedings of the 41st IEEE International Conference on Data Engineering. url: https://doi.org/10.1109/ICDE65448.2025.00050

join verification and applies a greedy algorithm to maximize additional data-point coverage over the regions already covered by the query dataset. CJSP similarly uses an R-tree for retrieval and additionally introduces the notion of indirect joinability, applying a greedy algorithm to maximize the global spatial coverage of the retrieved dataset collection, though at the cost of additional computational overhead. Both schemes operate entirely in the plaintext domain and therefore cannot be directly compared with PQJS under a unified encrypted evaluation protocol. Furthermore, their search objectives differ from ours: PQJS focuses on density-aware joinability evaluation rather than greedy coverage maximization. It is worth noting that while PQJS also employs MBR-based filtering in its pre-filtering stage, the subsequent ciphertext-domain vector computation over JCR and RER effectively resolves any false positives introduced by MBR intersection testing, ensuring that the final ranking is not affected by MBR approximation errors.

3 Detailed Dataset Statistics

Fig. 1 shows the spatial distribution of three real-world spatial data repositories used in the experiments. Table 1 provides basic information for each repository, where 'Points' indicates the total number of points in the spatial data warehouse, and 'Ave. Points' represents the average number of spatial location points in the datasets within the repository.

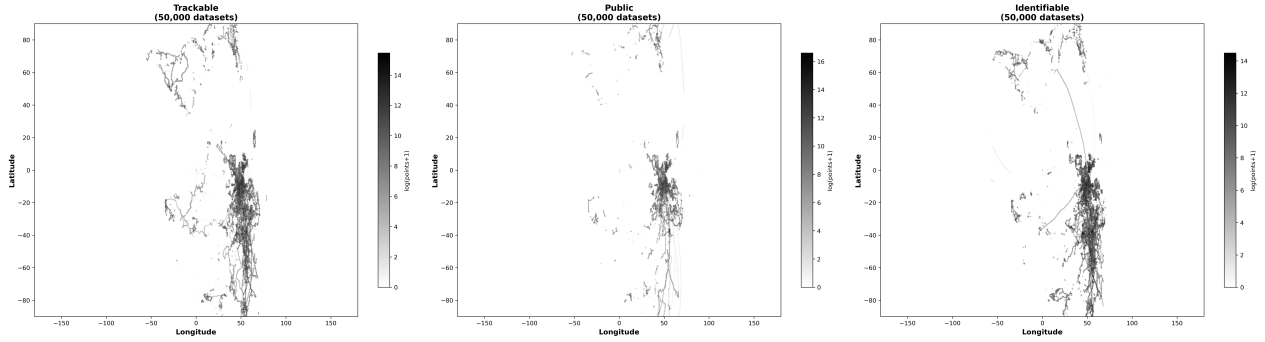


Figure 1: Spatial Data Repositories Spatial Distribution.

Table 1: Spatial Data Repositories Statistics

Repository	Size	Points	Ave. Points	Longitude	Latitude
Trackable	3.78GB	160,159,120	3,203	[-54.86, 78.31]	[-89.99, 90.00]
Public	2.20GB	102,568,677	2,051	[-53.16, 90.00]	[-89.98, 89.99]
Identifiable	3.99GB	158,785,717	3,175	[-61.51, 77.49]	[-89.99, 90.00]

4 Bloom Filter False-Positive Analysis

To evaluate the potential impact of Bloom filter collisions on the vectorized spatial representation, we further analyze the false-positive rate (FPR) under different Bloom filter configurations. The evaluation considers three spatial repositories, Trackable, Public, and Identifiable, with data scales ranging from $10K$ to $50K$. The Bloom filter dimension α is varied from 512 to 4096, while the number of hash functions γ is set to 3, 5, and 10. The theoretical FPR is estimated as

$$FPR \approx \left(1 - e^{-\gamma n/\alpha}\right)^{\gamma},$$

where n denotes the number of non-empty grids inserted into the Bloom filter.

Table 2 show that the false-positive rate is primarily determined by the ratio between the number of inserted non-empty grids and the Bloom filter dimension, rather than by the total number of spatial data

Table 2: Bloom Filter False-Positive Analysis

Repository	α	Number of Datasets								
		1w			3w			5w		
		γ			γ			γ		
		3	5	10	3	5	10	3	5	10
Trackable	512	6.49e-05	1.26e-06	1.16e-09	6.49e-05	1.26e-06	1.16e-09	6.49e-05	1.26e-06	1.16e-09
	1024	8.36e-06	4.28e-08	1.59e-12	8.36e-06	4.28e-08	1.59e-12	8.36e-06	4.28e-08	1.59e-12
	2048	1.06e-06	1.40e-09	1.84e-15	1.06e-06	1.40e-09	1.84e-15	1.06e-06	1.40e-09	1.84e-15
	4096	1.34e-07	4.46e-11	1.95e-18	1.34e-07	4.46e-11	1.95e-18	1.34e-07	4.46e-11	1.95e-18
Public	512	2.41e-05	2.46e-07	4.86e-11	2.41e-05	2.46e-07	4.86e-11	2.41e-05	2.46e-07	4.86e-11
	1024	3.07e-06	8.16e-09	6.04e-14	3.07e-06	8.16e-09	6.04e-14	3.07e-06	8.16e-09	6.04e-14
	2048	3.89e-07	2.63e-10	6.66e-17	3.89e-07	2.63e-10	6.66e-17	3.89e-07	2.63e-10	6.66e-17
	4096	4.88e-08	8.34e-12	6.91e-20	4.88e-08	8.34e-12	6.91e-20	4.88e-08	8.34e-12	6.91e-20
Identifiable	512	8.51e-03	3.19e-03	1.86e-03	8.51e-03	3.19e-03	1.86e-03	8.51e-03	3.19e-03	1.86e-03
	1024	1.26e-03	1.57e-04	1.02e-05	1.26e-03	1.57e-04	1.02e-05	1.26e-03	1.57e-04	1.02e-05
	2048	1.71e-04	6.18e-06	2.46e-08	1.71e-04	6.18e-06	2.46e-08	1.71e-04	6.18e-06	2.46e-08
	4096	2.23e-05	2.37e-07	3.82e-11	2.23e-05	2.37e-07	3.82e-11	2.23e-05	2.37e-07	3.82e-11

points. In particular, although the repository size increases from $10K$ to $50K$, the number of non-empty grids remains unchanged for each evaluated dataset, resulting in identical FPR values across different data scales. For datasets of Trackable and Public, the average numbers of non-empty grids are only 13 and 5, respectively, whereas datasets of Identifiable contains 39 average non-empty grids. Consequently, datasets of Identifiable exhibits a noticeably higher FPR under the same Bloom filter configuration. For example, with $\alpha = 512$ and $\gamma = 3$, the FPRs are 6.49×10^{-5}, 2.41×10^{-5}, and 8.51×10^{-3} for spatial datasets of Trackable, Public, and Identifiable, respectively. This observation confirms that the collision probability is closely related to the number of encoded non-empty grids rather than the raw number of data points.

Increasing the Bloom filter dimension substantially reduces the FPR. For datasets of Identifiable with $\gamma = 3$, increasing α from 512 to 4096 reduces the FPR from 8.51×10^{-3} to 2.23×10^{-5}, corresponding to a reduction of more than two orders of magnitude. Similar trends are observed for datasets of Trackable and Public. Therefore, a sufficiently large vector dimension provides a sparse representation and effectively mitigates Bloom filter collisions. The number of hash functions also affects the FPR, although its effect is less pronounced than that of the Bloom filter dimension in the evaluated configurations. Increasing γ from 3 to 10 consistently reduces the FPR across the tested datasets and dimensions. For example, for datasets of Identifiable with $\alpha = 512$, the FPR decreases from 8.51×10^{-3} at $\gamma = 3$ to 1.86×10^{-3} at $\gamma = 10$. These results indicate that increasing the number of hash functions can further reduce the collision probability under the evaluated parameter range. Overall, the measured FPR remains low under the parameter settings adopted in PQJS. In particular, when $\alpha = 4096$, the FPR is below 2.23×10^{-5} for datasets of Identifiable with $\gamma = 3$ and further decreases to 3.82×10^{-11} with $\gamma = 10$. For datasets of Trackable and Public, the corresponding FPRs are even lower. These results suggest that Bloom filter collisions are unlikely to introduce substantial false-positive effects in the evaluated parameter range.

5 Detailed Time Breakdown

Fig. 2 illustrates the detailed time breakdown of the PQJS search process. The results show that CKKS-based encryption time incurs the largest time overhead, followed by GMBR-based filtering. It is worth noting that the encryption time represents the one-time offline cost of encrypting all datasets in the spatial repository, rather than a per-query overhead. The GMBR-based filtering stage involves computing the GMBRs for all datasets in the repository and the exemplar dataset $\mathcal{D}_e$, together with one round of communication and decryption. Despite its non-negligible overhead, GMBR filtering substantially reduces the number of candidate datasets, thereby greatly alleviating the subsequent homomorphic computation time. In contrast, token generation time and the final decryption and sorting stage each account for less than 1% of the total

search time, where the decryption cost corresponds only to the top-k result datasets returned by the search rather than the entire repository. These results indicate that their impact on the overall search efficiency is negligible.

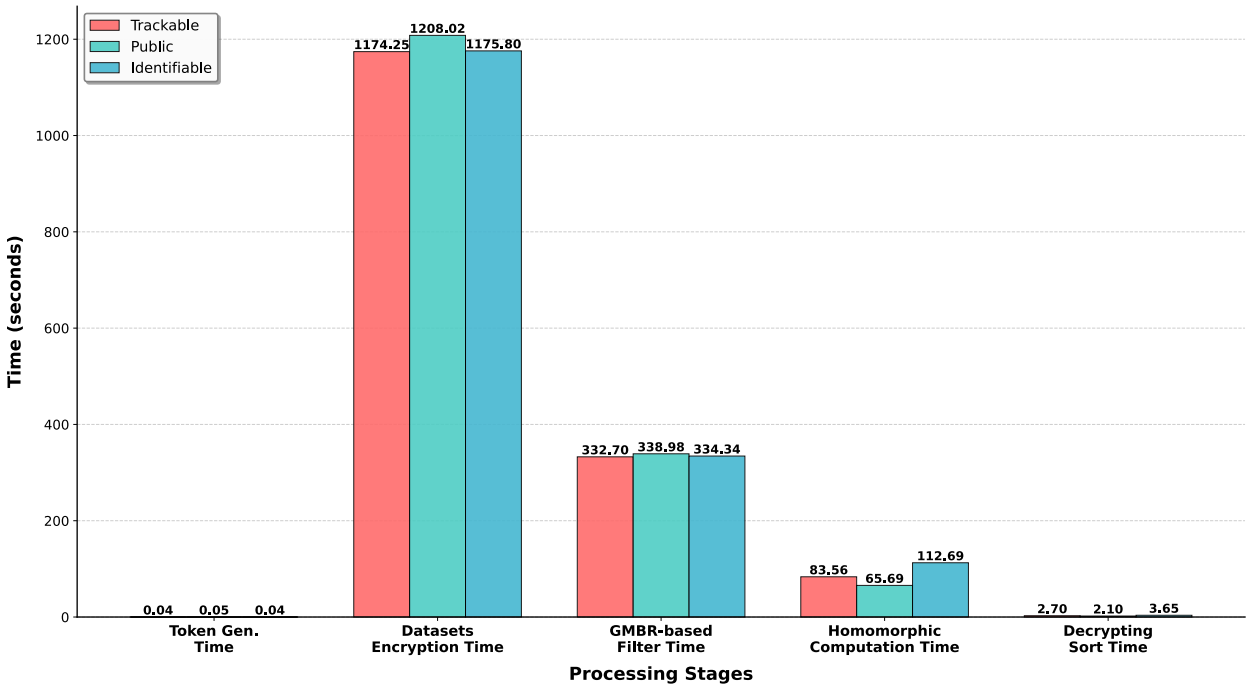


Figure 2: Detailed Time Breakdown of Search Processing.

6 Spatial Search Result Visualization

Fig. 3 shows the actual search results of the PQJS scheme over the three spatial data repositories. The red box represents the spatial range of the exemplar dataset, while the grayscale regions indicate the returned search results, with darker shades corresponding to higher data-point densities. In PQJS, SAU is defined as the product of JCR and RER, which provides a balanced treatment of the two factors without introducing an additional weighting parameter. This formulation effectively assigns equal importance to spatial connectivity and density-weighted spatial augmentation. Nevertheless, the weighting between these two factors can be flexibly adjusted according to different application requirements. Specifically, SAU can be generalized as $SAU = \alpha JCR + (1-\alpha)RER$, where $\alpha \in [0,1]$ controls the relative importance of spatial connectivity and spatial augmentation.

7 Scalability and Updateability Analysis

As the number of spatial datasets in the repository increases, the GMBR-based pre-filtering strategy maintains a high filtering rate and becomes increasingly beneficial for large-scale repositories. In particular, introducing more spatial datasets provides a larger search space in which spatially disconnected datasets can be efficiently pruned before the subsequent homomorphic computations. Therefore, the proposed PQJS scheme can effectively reduce the number of candidate datasets requiring expensive ciphertext-domain evaluation, demonstrating its scalability for large spatial repositories.

In addition to scalability, PQJS supports flexible updates to the spatial data repository. The addition or deletion of a dataset does not require rebuilding or modifying the representations of other datasets, since the spatial representation and ciphertexts of each dataset are generated independently and are not dependent on other datasets or the exemplar dataset. When a new dataset is added, its spatial information

is mapped to the corresponding vector representation and encrypted using the public key before being inserted into the encrypted repository. When a dataset is removed, its corresponding ciphertext vector set can be directly deleted without affecting the remaining encrypted datasets or the search token generation process. This independent representation and encryption mechanism provides PQJS with robustness against dynamic dataset insertion and deletion, making it suitable for continuously evolving spatial data repositories.

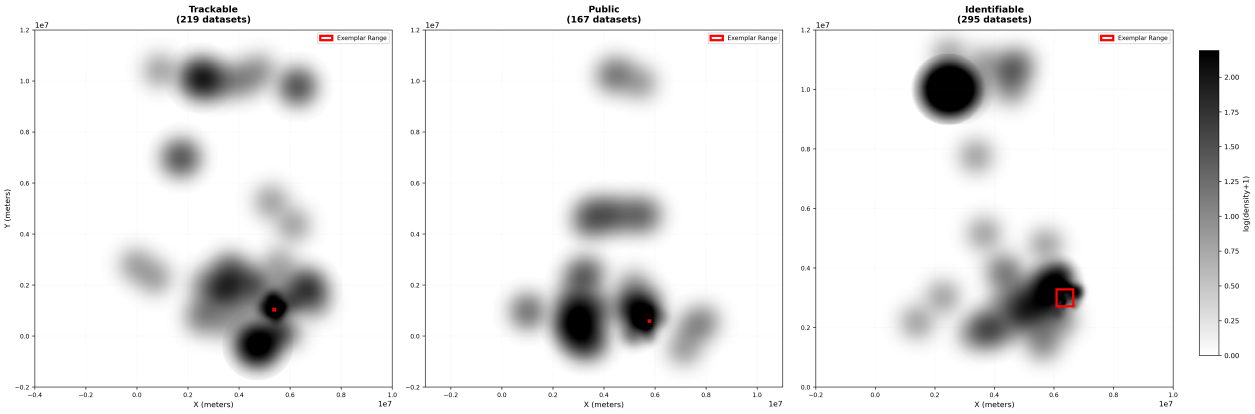


Figure 3: Spatial Search Result Visualization.

8 Limitations and Future Work

Despite the effectiveness of PQJS in enabling privacy-preserving spatial dataset joinable search, several limitations remain to be addressed. First, the security of the proposed dual-server architecture relies on the non-collusion assumption between the semi-honest SS and AS. If the two servers collude, the jointly available ciphertexts, evaluation information, and decryption capability may lead to additional privacy leakage. Developing a collusion-resistant multi-server or threshold-decryption architecture is therefore an important direction for future work.

Second, although the proposed GMBR-based pre-filtering strategy substantially reduces the number of candidate datasets and avoids unnecessary homomorphic computations, CKKS-based operations still introduce considerable computational and storage overhead compared with plaintext or conventional symmetric-key solutions. In particular, the initial ciphertext generation and GMBR filtering constitute the major computational costs of the current implementation. Future work will investigate more lightweight homomorphic representations and optimized ciphertext evaluation techniques to further reduce the overhead while preserving the required security guarantees.

Third, the vectorized spatial representation relies on Bloom filters to encode non-empty spatial grids, which inevitably introduces a small probability of false positives due to hash collisions. Although our experimental analysis shows that the false-positive rate remains low under the adopted parameter settings, a probabilistic representation may still introduce minor deviations in spatial matching. Future work will explore collision-resistant or adaptive spatial encoding methods to further improve the accuracy of spatial joinability evaluation.

Finally, the current experimental evaluation is conducted on spatial repositories with up to tens of thousands of datasets. Although the scalability experiments demonstrate the effectiveness of GMBR filtering in reducing the candidate set, further evaluation on million-scale spatial repositories and more diverse spatial distributions is still necessary to fully characterize the scalability of PQJS in large-scale real-world deployments. Future work will therefore focus on larger repositories, more heterogeneous spatial distributions, and adaptive filtering strategies for large-scale spatial data markets.